

Q341-Q396

From the analytic contract to the first certified historical brackets

Horizon / Goldbach Program · Serge Durand

Continuation synthesis · September 9, 2026

The analytic chain now connects the primary auxiliary to zeta on the entire critical line. The remainder has an effective bound, eight atoms are certified, and six historical brackets yield genuine distinct zeros. This volume traces the constructions, changes of route, and contracts that remain open.

8 / 8

atoms at the canonical point

6 / 649

closed historical brackets

Bridge A: OPEN. Latest documented closeout: Q395.

Q396: continuation mandate; results not supplied.

No unconditional proof of Goldbach is claimed.

EXECUTIVE SYNTHESIS

Where do we stand?

At the latest documented closeout, Q395, the project has certified six historical brackets below height 1000. Each has two strict signs of the true Hardy function and contains a genuine zero of riemannZeta. Since the intervals are disjoint, the six zeros are distinct.

Level	Documented status as of September 9, 2026
Analytic transport	Auxiliary-zeta identity on the entire critical line and canonical Riemann-Siegel decomposition: obtained.
Analytic remainder	Specialized bound of 2.5×10^{-30} , followed by a uniform bound of 5×10^{-30} on [685, 700.3]: obtained on their respective domains.
Pointwise computation	Eight real atoms, membership in a rational box, and a strict sign: obtained at the Q393 canonical point, then at the new Q394-Q395 points.
Historical family	6 closed rows out of 649: brackets 410-415, Lean indices 409-414. These results do not certify the other 643 rows.
Counting	Multiplicity at least 6; at least 7 with the separate interval (14,15). No exact global count follows.
Q396	Available mandate: optimize the coefficient recurrence and target 405-409. No results report was supplied.

Preserved frontier: BRIDGE_A = OPEN; TS340_UNCONDITIONAL = OPEN_FROZEN. This continuation claims neither an unconditional proof of Goldbach nor a proof of the Riemann hypothesis.

The major change since TS340 is concrete: the project now has a chain that starts from independently defined analytic functions, verifies rational computations, and then produces localized zeros. Extending this subfamily to an exhaustive coverage remains to be proved.

VOLUME OUTLINE

Reading guide and scope

Pages	Contents
4-5	From the TS340 contract to the $H = 1000$ pilot; Q341-Q346 and Hardy conventions.
6-7	Q347-Q360: rational computation, theta-Mellin quadrature, and the first robust sign.
8-11	Q361-Q380: the entire function, coefficients, independent remainder, and contour proof.
12-14	Q381-Q389: Mordell, theta-Mellin, and transport to the entire critical line.
15-17	Q390-Q395: an effective remainder bound, eight atoms, and six historical brackets.
18-20	Open contracts, the Q396 program, and the scope of the audits.
21-26	Chronological record: every milestone from Q341 to Q396.
27-28	Notation, reference materials, and reproducibility landmarks.

How to read the statuses

Proved denotes a result reported as compiled, or a declaration found in the supplied source chain. **Conditional** denotes a theorem that still requires a semantic premise. **Diagnostic** denotes an external computation or a performance measurement. **Mandate** denotes authorized work; authorization alone does not imply a completed result.

The chronology distinguishes a campaign verdict from its later continuations. For example, an object left open in Q376 can be connected in Q379; this does not retroactively turn the initial report into a complete success.

Documentary limit. The requested numbering begins with TS341, the permanent result on zero ordinate. Q342, Q345, Q355, Q359, and Q360 are partly documented by later references; Q371-Q372 explicitly remained at the gates of their mandate. Q396 is described as work to execute or document, rather than as an achieved closeout.

This volume synthesizes the supplied reports, sources, and attestations. Its preparation did not launch a new Lean compilation of the entire historical codebase. References [S1] to [S13] are detailed on the final page.

FROM TS340 TO BRIDGE A

The original contract and the pilot

The TS1-TS340 synthesis described a conditional spectral-budget architecture. Explicit formulas, infinite tails, and adapters were available; certificates counting genuine zeros remained to be constructed. TS341 then discharged the obligation excluding a zero of ordinate zero in the relevant contract.

Scale	What it requires	Current status
TS340 reference	Height 1,132,490; exact positive count 2,001,050; disjoint local counts and saturation.	Historical contract remains open.
Q343 pilot	Height 1000; candidate family of 649 brackets; two Turing bounds at level 1001.	Geometry obtained; semantic subfamily of six rows.
Q393 computation	One rational endpoint of historical bracket 415.	Value enclosed and positive sign proved.

Success in a pilot below 1000 therefore does not establish the count required by TS340 at its reference height. It provides a method and initial certificates that can support subsequent work.

Three connections that must remain distinct

The early audits distinguish connection **A**, which gives semantic meaning to evaluations and numerical boxes, from connection **B**, which identifies Hardy phase conventions. The transfer from the Hardy sign to the carrier, then to the project's xi function, also has mathematical theorems. Q346 closes B without certifying a FLINT execution.

Recent reports also use “Bridge A” as the overall status of the certification campaign. In this volume, evaluation, family, Turing, and saturation obligations remain separately identified: closing one endpoint does not close all of these contracts.

The candidate datum “649” remains data until the signs, counting upper bounds, and bindings to the correct payload have proofs. A list length is not a count of all zeros.

[S1, S2, S11, S12]

Q341-Q346

Defining the right function to certify

TS341: removing an old premise

The alternating eta series, grouped into pairs, is connected to its real limit and to Mathlib's zeta conventions. The terminal result `noZeroOrdnateInTruncation` closes the zero-ordinate obligation. The first volume mentioned it outside its scope; here it forms the starting point.

Q342-Q344: from data to contracts

Q343 reconstructs Q342 connections: a sign change yields a zero on the critical line; disjoint brackets yield a multiplicity lower bound; a global upper bound and saturated lower bounds yield an exact count. The standalone Q342 report is not among the materials consulted.

The Q343 pilot produces 649 dyadic brackets and checks their data with FLINT at several precisions. Its initial structural replay uses `native_decide` and `Lean.ofReduceBool`; it is explicitly excluded from the analytic proof of signs. The semantic theorems are audited with the three standard axioms.

Q344 fixes the two rationals of the first serialized bracket and states membership obligations for the corresponding boxes. It distinguishes the sign of a number contained in a box from the proof that the box actually contains Hardy Z.

Q345-Q346: phase and logarithm branch

Q346 records that the announced durable Q345 package and manifest are missing; this documentary gap is preserved. It nevertheless establishes the mathematical convention binding from pinned FLINT sources:

```
exp(i · (Im L - t log π / 2)) = ΓR(1/2 + it) / |ΓR(1/2 + it)|
provided that exp(L) = Γ((1/2 + it)/2).
```

Equality of the exponentiated phase is independent of the valid choice of logarithm. The documented Hardy formula is then identified with `normalizedHardyZ`. No numerical endpoint sign has yet been obtained.

Q347-Q354

A verifier alone is not yet enough

Q347 constructs the small rational computation language that will remain central to later certificates. Its soundness says: if the computation is accepted and every atom belongs to its box, then the output box contains the expression's value. The second condition is the decisive analytic obligation.

```
check(atomBox, cert) = true
+  $\forall n, \text{atomBox}(n).\text{mem}(\text{atomValue}(n))$ 
 $\Rightarrow \text{cert.output.mem}(\text{cert.expr.value}(\text{atomValue}))$ 
```

The first Q347 attempt stops because the environment is unavailable; the retry compiles the kernel outside its strict window. Q348 then repairs the complete binding and replaces the demand for equality of boxes with the correct inclusion direction: the proved box must refine the serialized box.

Milestone	Analytic or computational progress
Q348	Exact reduction of completed zeta to a theta-Mellin integral, with polar term $1/(t^2 + 1/4)$.
Q349-Q350	Exponential change of variable, folding onto the half-line, and truncation to $[0,2]$; effective tail $\leq 64/70336337001 < 10^{-9}$.
Q351-Q352	Trapezoidal-rule error under a C^2 hypothesis; second derivative ≤ 25 for $ t \leq 15$. At 8192 panels, total error $\leq 1/4,000,000$.
Q353	Trace language and semantic invariant proved. The first concrete replay reaches an elaboration limit.
Q354	Flattened instructions and registers, short blocks, and cosine Taylor expansion. One analytic node at $u = 1$ is certified for $t = 14$ and $t = 15$.

A proof for one quadrature node is not yet a proof for the whole sum. Q354 certifies neither the sign of the integral nor the first serialized bracket.

Q355-Q360

From quadrature to the first zero

Q355 appears in references to flat interval kernels; its standalone closeout report was not found. Q356 and Q357, however, document the scaling effort and its binding to the true integrand.

Stage	Effective coverage	Limit at the checkpoint
Q356	17 nodes, 16 panels; signed outward dyadic rounding, parametric Taylor bounds, and a uniform theta tail.	Standalone pilot compiled. Nominal Q349 binding not replayed in this environment.
Q357	129 nodes, 128 panels; independent blocks and balanced aggregation.	Binding to the canonical Q349 integrand closed; the complete sum remains to be built.
Q358	Sum of all 8192 panels bound to the trapezoidal rule; quadrature leaves and signs at 14 and 15.	A genuine zero in (14,15). This broad interval is not the first narrow ledger bracket.

```
normalizedHardyZ 14 < 0    and    0 < normalizedHardyZ 15
⇒ ∃ t ∈ (14,15), riemannZeta(1/2 + it) = 0
```

This first route obtains robust signs at low height without waiting for the full Riemann-Siegel theory. It will be reused as a separate interval in the Q394-Q395 lower bounds.

The pivot to Riemann-Siegel

The Q359-Q360 diagnostics investigate another route for endpoints much closer to zeros. Q361 retains an essential correction: a derivative capacity limited to indices 0-9 only supports $K \leq 3$, whereas order K requires derivatives through $3K$. The older $K = 4-30$ experiments remain incomplete.

Q360 supports numerical routing and target selection; it does not replace a proof of the formula or its remainder. The project must now build the analytic link connecting the finite sum to the true Hardy Z.

[S4, S5]

Q361-Q365

Making the coefficients mathematically sound

Q361 fixes the parameters $a(t) = \sqrt{t/(2\pi)}$, $N(t) = \lfloor a(t) \rfloor$, and $p(t) = 1 + 2(N(t) - a(t))$, then the elementary properties of the error envelope. Its audit shows that the Riemann-Siegel formula and its analytic remainder are not already available in the pinned revision.

Q362: the primary formula and its singular points

The published quotient has denominator $2 \cos(\pi z)$, which vanishes at half-integers. An incorrect preliminary candidate is rejected after reading the primary source. Lean proves the denominator's zero set, simplicity of its zeros, and the first common zero of the numerator.

$$\begin{aligned} N(z) &= \exp(\pi i(z^2/2 + 3/8)) - i\sqrt{2} \cos(\pi z/2) \\ D(z) &= 2 \cos(\pi z) \\ F(z) &= N(z)/D(z), \text{ away from half-integers.} \end{aligned}$$

Q363-Q364: a genuine entire function

Q363 handles all half-integers by phase reduction modulo 4. Q364 constructs the local extension through divided differences (dslope), proves local nonvanishing of the regularized denominator, and then glues an entire function.

At a singular point of the raw quotient, F takes the value N'/D' . Agreement with the quotient elsewhere and global holomorphy then justify iterated derivatives. A default derivative value is not used as an analytic proof.

Q365: exact recurrences and indices

The d coefficients, their initial conditions, support, and exceptional top recurrence are formalized. The C_k are connected to derivatives of F ; the maximum order is $3K$. The sum through K is exact, but it has not yet been identified with part of zeta.

At $K = 63$, this interface will require 190 jets, of orders 0 through 189. Their analytic existence is established here; their numerical boxes will only be certified in Q393.

Q366-Q372

Defining the remainder before bounding it

The first Q366 report closes with its provenance gate unmet: the independent definition of the remainder was not sufficiently accessible. This decision avoids replacing the remainder with a simple difference between the target and the approximation.

The Q366-R continuation then retains a 2024 source and genuinely defines the integrals `sourceRgK` and `sourceRSRemainder`, with directions, principal logarithm, and orientation conventions. The decomposition still remains an explicit contract to prove.

Milestone	What is available	What is still missing
Q367	Algebraic transport of a raw bound to Hardy normalization.	The required raw bound is a premise, rather than a constructed witness.
Q368	Finite-sum certificate enlarged by a rational radius; checker soundness.	Atom values, the decomposition, and a bound for the true remainder.
Q369	Affine geometry, avoidance of poles and the branch cut, inner integrability, and truncated Cauchy rectangles.	Complete domination, Fubini, and the infinite contour shift retain their analytic gates.
Q370	Exponent factorization and separable integrable majorants; a uniform cosine lower bound.	A strict harmonic gap, followed by suitable quantitative constants.
Q371-Q372	Planned production endpoint and bracket targets.	Not attempted at this stage: upstream analytic gates remain unmet.

The logarithmic function `sourceF` in this remainder is not Q364's entire function `F`. Similar names do not justify replacing one function with the other.

The product of two integrable majorants permits Fubini only once its witnesses have been constructed. The chronology follows these obligations one by one, without promoting a conditional framework alone to a canonical formula.

[S6]

Q373-Q376

Domination and contour shifting

Q373: closing the harmonic gap

With $H = 1 + \text{sourceF}$ and $G = \exp(-H)$, Lean establishes boundary extensions, positivity of $\text{Re}(H)$ in both half-planes, and then a uniform gap on admissible lines. A compact part and an explicit tail are combined to obtain a strictly positive a .

The Q370 gap witness is now constructed. Together with the cosine bound, the double kernel has an integrable majorant, and the Fubini interchange no longer has a residual harmonic premise.

Q374-Q375: the proper scope of the first bounds

Q374 exposes Fubini for the source remainder and distinguishes the 2011 and 2024 expansion conventions. Q375 bounds the remainder norm by the product of majorant integrals; continuations produce closed forms of these bounds. A bound of this kind is not yet the numerical Arias bound required at an endpoint.

Q376: handling the rectangle sides

Branches, poles, tangents, and orientations are checked in the central, lower, and upper regimes. Gaussian majorants make the transverse sides vanish for the regular primary function and for finite polynomials. Cauchy on truncated rectangles then permits shifting between complete lines.

```
regular primary function - finite Taylor polynomial
?= sourceRgK on the final line
```

The pointwise identity above remains the open leaf. A Cauchy representation valid inside a local disk does not automatically extend to an entire unbounded line. Q377-Q379 will resolve precisely this binding.

Two quantitative levels remain distinct: the existence of a positive gap, useful for analysis, and an explicit rational gap sharp enough to certify a particular numerical margin.

Q377-Q380

The source representation becomes a theorem

Milestone	Obstacle actually addressed
Q377	Analyticity in the parameter and an open, preconnected transport domain. The global binding awaits an independent local identity.
Q378	Cauchy representation on an annulus; transport of the outer circle. The inner contour is parametrized by a Blaschke transformation.
Q379	Global angular lift, degree 1, an increment of 2π , and oriented substitution. The local identity feeds continuation and then the Q376 deformation.
Q380	Independent definition of $R(s)$ by its meromorphic contour; integer poles, one-pole jumps, telescoping, and saddle-point substitution.

The Q380 shift removes a pole's principal part, controls the holomorphic remainder, and sends the transverse sides to zero. The exact $2\pi i$ jump supplies the Dirichlet term; a finite sum of shifts produces the full prefix.

$$R(s) = \text{Dirichlet prefix} \\ + \text{prefactor} \times (\text{primary finite correction} + \text{source remainder})$$

This identity concerns an auxiliary function defined before the formula. The source-level Riemann-Siegel formula is therefore closed, rather than made true by a circular definition of the remainder.

What this success does not yet provide

The independent relation between this $R(s)$ and riemannZeta is still missing at the Q380 closeout. Transport to Hardy Z , followed by sign certificates, therefore cannot yet be claimed.

The Q380 report preserves its original chronological verdict, **OVERRUN**. Results obtained during the explicitly authorized continuation are classified **POST_DEADLINE_USER_AUTHORIZED**; technical success does not rewrite that chronology.

[S7]

Q381-Q383

Building the auxiliary-zeta anchor

Q381: conventions and analyticity

The Γ_R factor is fixed exactly to Mathlib's convention. Conjugation of the reflected contour includes the logarithm branch, the denominator sign, and parameter reversal. The primary auxiliary function is proved entire by differentiation under its integral, using a local Gaussian majorant.

```
Rreflected(s) = conj(R(1 - conj(s)))  
s = 1/2 + it  ⇒  1 - conj(s) = s
```

On the critical line, the completed auxiliary pair reduces to twice a real part. The Hardy connector remains conditional on the auxiliary-zeta identity.

Q382: the primary Mordell formula

The closed form of the Mordell integral is proved without using zeta: contour shifting, the pole, lateral decay, translations, completing the square, and the exact complex Gaussian integral. This identity supplies the analytic anchor that symmetry arguments alone lacked.

Q383: the theta-Mellin side

The regularized Mordell kernel is extended at zero with value $-1/2$. On a nonempty open Mellin domain, Lean proves theta-line integrability, absolute domination of its series, the sum-integral interchange, and identification of the Gaussian Mellin transform with completed zeta.

Established side	Binding still needed at this stage
Primary Mordell integral: exact formula.	Transport from the auxiliary contour to the source theta representation.
Gaussian Mellin integral: completed zeta.	Actual construction of an inhabitant of <code>SourceAuxiliaryThetaSeed</code> .

A functional equation or identity theorem can propagate an equality already established on a suitable domain. It does not, by itself, provide that initial equality.

Q384-Q388

Exceptional points: a useful correction

Q384 closes transport on a regular strip and on the seed domain with a discrete set removed. The difficulty arises at positive odd points, where the reflected Γ_R factor, as totalized in Mathlib, equals zero.

$$\Gamma_R(1 - \text{conj}(s)) = 0 \Leftrightarrow \exists n : \text{Nat}, s = 1 + 2n$$

The initial global connector then requires a completed theta expression to vanish at negative even points. The following campaigns show why this premise cannot be mechanically deduced from a vanishing inverse Gamma factor.

Milestone	Result actually obtained
Q384	Regular transport and exact characterization of the exceptions. The exceptional-point connector retains a premise.
Q385	Γ_R and its totalized inverse equal zero at $-2(n+1)$. The uncompleted entire auxiliary function vanishes there.
Q386	The completed theta expression is regular at these points and equals exactly one half of a line integral.
Q387	If this value is nonzero, the old Gaussian seed at the positive odd point cannot hold. Nonvanishing itself is not proved.
Q388	Separation of the constant mode; an exact zero criterion through cancellation with the residual integral.

$$\begin{aligned} \text{Exceptional value} &= (\text{residual integral} - (2/s) \cdot I^{(s/2)})/2 \\ \text{Value} = 0 &\Leftrightarrow \text{residual integral} = (2/s) \cdot I^{(s/2)} \end{aligned}$$

Neither a closed-form value nor a nonvanishing proof for this integral is obtained in Q385-Q388. The completed theta auxiliary function must not be identified with completed zeta without proof. Q389 will bypass this detour for the critical line.

Q389 / RS17H

Transport across the entire critical line

Q389 closes the relation needed for Hardy computations for every real t . The old global seed at exceptional points leaves this proof's critical path; it is not artificially instantiated.

```
sourceAuxiliaryPair (criticalPoint t)
  = sourceCompletedZeta (criticalPoint t)

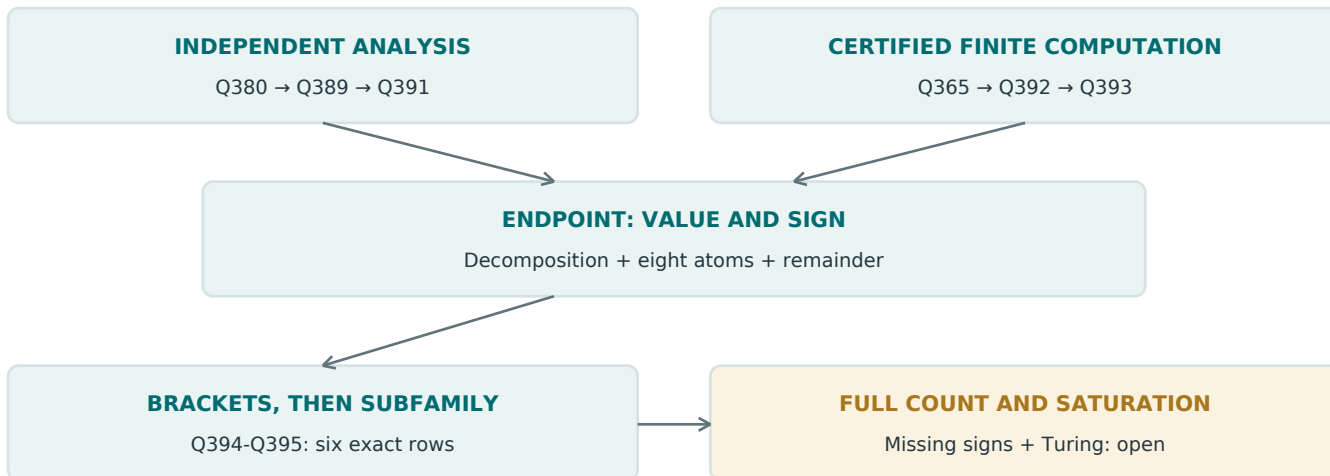
normalizedHardyZ t
  = sourceNormalizedAuxiliaryCarrier t
```

Two regular domains, then height zero

The domains $0 < \text{Re}(s) < 2$ with $\text{Im}(s) > 0$, or $\text{Im}(s) < 0$, are open and preconnected. The proper analytic representatives satisfy the identity obtained from the regular seed there. Nonvanishing of Gamma off the real axis identifies them with the historical expressions.

At $t = 0$, a sequence of positive heights $1/(n+1)$ approaches the critical point. Continuity and uniqueness of limits give the remaining case. A trichotomy on t thus covers the entire critical line.

The reusable cascade



The source formula becomes an exact decomposition of Hardy into a finite part and an independent remainder contribution. The Q368 checker and a generic `AnalyticLeaf` constructor are connected. A concrete certificate still requires atom boxes and a proved rational radius for the true remainder.

[S9, S10]

Q390-Q391

The remainder becomes sufficiently small

The canonical point is fixed at $t_0 = 751937898017/1073741824$, with $K = 63$ and $\text{ell} = 10$. Q390 proves $\text{rsN}(t_0) = 10$, the central regime, the geometry of ξ and q , a rational cosine bound $M = 1$, and the exact decomposition at this point.

Q390: envelope arithmetic is not enough

Lean establishes $\text{explicitAriasBound}(t_0, 63) \leq 10^{-14}$. But no proof yet connects this number to the true source remainder: the proposition `RequestedAriasRemainderTransport` remains open. A diagnostic of the existing Q375 majorant indicates an envelope far too wide for the point's margin; this motivates a new estimate.

Q391: a specialized direct proof

The primary remainder is connected to the Taylor difference by the representation theorems already constructed. Order-64 cancellation controls the central part; exponential envelopes control the tails. All integrability statements and the identity with the source integrand are proved.

Contribution or binding	Bound obtained
Central integral on $[-17/2, 17/2]$	$\leq 1.5 \times 10^{-30}$
Sum of tail envelopes	$\leq 1 \times 10^{-30}$
Norm of the exact prefactor	$\leq 1/2$
True normalized Hardy remainder at t_0	$\leq 2.5 \times 10^{-30} = 1/4000000000000000000000000000$

```
Q391Probe.requestedAriasRemainderTransport :  
Q390Probe.RequestedAriasRemainderTransport
```

The terminal theorem has no argument or residual semantic premise. This specialized proof bounds the independent 2024 remainder and compares it with the requested envelope; it does not prove the generic Arias theorem for all points, or a general equality between the 2011 and 2024 remainders.

[S10]

Q392-Q393

Eight atoms and an unconditional endpoint

The canonical finite part is $2 \operatorname{Re}(\mathbf{G} \times (\mathbf{S} + \mathbf{A} \times \mathbf{B}))$. The four complex numbers G , S , A , and B supply eight real values. A positive rational computation becomes a proof of the Hardy sign only after all eight values have been certified.

Atoms	Exact object	Closure
0-1	Normalized Gamma factor: real and imaginary parts.	Q392, Gamma/Dirichlet checkpoint.
2-3	Dirichlet prefix: real and imaginary parts.	Q392, the same checkpoint.
4-5	Canonical transform prefactor.	Q392: widths $\leq 10^{-17}$.
6-7	Finite correction, with the primary phase and Q365 coefficients evaluated at $-\mathbf{x}_i$ and $\mathbf{q}$.	Q393: two boxes of width 10^{-20} .

Q392 stabilizes the prefactor through an exact identity involving exponentials, logarithms, and arctangent, then certifies its rational leaves. The remaining premise concerns only the two components of B .

Q393 uses the global identity $D \cdot F = N$ to construct a triangular recurrence of normalized jets. The 190 quotient jets, 64 coefficient rows, and 3072 assembly contributions are bound to the true analytic objects; the derivative factorials are restored.

```
requestedEndpoint_mem      : certified box contains Hardy Z(t₀)
requestedEndpoint_positive : 0 < Hardy Z(t₀)
requestedEndpointAnalyticLeaf : concrete AnalyticLeaf at rational t₀
```

The exact output is summarized in decimal as $[1.8855174438 \times 10^{-13}, 1.8915581358 \times 10^{-13}]$, with width approximately 6.041×10^{-16} . Lean uses only its rational endpoints and the Q391 radius of 2.5×10^{-30} .

This point is the right endpoint of historical bracket 415. Its positivity certifies neither the left endpoint, a much narrower old serialized box, nor the entire family.

Q394-Q395

Six ledger rows contain genuine zeros

Q394 independently reconstructs the eight atoms at three new points: the left endpoint of 415 and both endpoints of 414. The Q393 endpoint supplies the remaining right endpoint. The two brackets yield two genuinely localized, distinct zeros. Neither uniqueness nor simplicity is inferred from a sign change alone.

Historical bracket	Exact left endpoint	Exact right endpoint	Signs
414	3002749485653 / 4294967296	1501374742827 / 2147483648	+ / −
415	3007751592067 / 4294967296	751937898017 / 1073741824	− / +

A uniform remainder bound of 2.5×10^{-30} is constructed on [699, 700.3]. The separate Q394RangeExtended extension increases the radius to 5×10^{-30} on [685, 700.3] and covers the eleven historical spans 405-415.

Q395 uses this extension to certify eight new endpoints: four complete pairs, brackets 410-413. The new proofs are integrated into the historical subfamily with the same indices and the same rationals.



Historical rows 405-415: a remainder bound covers the entire range.
Only the six rows 410-415 have two certified strict signs here.

```
closedIndices = {409, 410, 411, 412, 413, 414} ⊂ Fin 649
closedIndices.card = 6
positiveMultiplicityMass 1000 ≥ 6
```

The Q358 interval (14,15), separate from this subfamily, raises the lower bound to 7 after disjointness is proved. It adds no ledger row. The geometry of all brackets is available, but five pairs in the covered batch and 643 rows in the global ledger still lack certified signs.

The final report, status, and HistoricalClosedFamily source supersede the figures 3 or 4 left in intermediate Q395 notes. The documentary correction does not rewrite the sealed archive.

OPEN CONTRACTS

What still separates the pilot from Bridge A

Obligation	Precise status
Other family signs	643 historical rows are uncertified; among them, 405-409 are already covered by the uniform remainder bound and await their finite computations.
Complete semantic family	H1000ScaledLambdaBracketFamily requires the true signs of all 649 rows. Geometry alone does not supply this sign function.
Two Turing upper bounds	initialTuringMultiplicityMass 1001 ≤ 288 and upperTuringMultiplicityMass 1001 ≤ 361: open.
Exact count	positiveMultiplicityMass 1000 = 649: not obtained. Local lower bounds alone do not imply it.
TS329 saturation	On one and the same payload: global count, local lower bounds, exact sum, disjointness, and cap compatibility.
First serialized bracket	Both old enclosure obligations for the first narrow interval remain open. The zero in (14,15) does not replace them.
Serialized refinement	Membership in a new box and inclusion in an old box are two different theorems.
TS340 reference contract	Height 1,132,490 and the 2,001,050 zeros of the original contract remain outside the current pilot closeout.

A precise width obstruction

A symmetric inflation of radius 2.5×10^{-30} has width at least 5×10^{-30} . It therefore cannot refine the historical right box of 415, whose width is $1572/10^{80}$. Q394 proves this obstruction for this strategy; it does not refute mathematical membership of the true number in the old box.

The direct sign route can construct the family without requiring this serialized refinement. It does not thereby prove the refinement. Carrier signs are transported by a positive scale; Hardy numerical boxes do not become carrier boxes.

[S2, S11, S12]

Q396 / RS24H: DOCUMENTED MANDATE

Next step: a reciprocal kernel and the 405-409 batch

Status at the date of this synthesis: a Q396 mandate is available; no Q396 report, closeout status, or results archive was supplied. The objectives below are therefore not counted as achieved.

Testing an improvement on a complete chain

The mandate proposes certifying a single box for the inverse of the denominator's constant term D_0 at q , then reusing that box at every step of the jet recurrence. The inverse is D_0^{-1} , not q^{-1} .

$$F_0 = N_0 \times D_0^{-1}$$

$$F_{n+1} = (N_{n+1} - \sum_{j=0..n} D_{j+1} \times F_{n-j}) \times D_0^{-1}$$

Nonvanishing of D_0 , membership of the true inverse in its box, and outward rounding must be proved. The experiment must cover all 190 jets, the canonical correction, and the final semantic root. A first and a last numerical block are not sufficient.

Closing pairs already covered analytically

The priority target consists of the ten exact endpoints of brackets 405-409, with $K = 63$, $\text{ell} = 10$, and the uniform radius of 5×10^{-30} . The first six values prepared at a point can only be reused with their true membership theorems at that same point.

If m new pairs close	Result to prove
$0 \leq m \leq 5$	Historical subfamily of cardinality $6 + m$; distinct zeros and corresponding lower bounds.
$m = 5$	11 exact rows, multiplicity at least 11; at least 12 with (14,15), separately.
After the complete batch	638 other rows would remain to be certified, together with the upper bounds and saturation.

The planned window is four hours, with a verification reserve. Adopting an optimization depends on a measured gain across the true chain; the older methods remain the fallback. Faster computation closes no analytic theorem outside the domain already proved.

TRUST AND REPRODUCIBILITY

What the audits actually attest

The final certificates rely on the standard axioms `propext`, `Classical.choice`, and `Quot.sound`. Recent audits traverse types and term constants to verify the correct atoms, endpoint, decomposition, and remainder, and the absence of a remaining semantic premise.

Campaign	Reported compilation scope	Package integrity
Q391	27/27 direct sources; 87 audited declarations.	81/81 manifest entries; 82/82 files in the ZIP replay.
Q392, prefactor	16/16 modules; 14 axiom audits for the checkpoint.	71/71 entries; previous archives preserved.
Q393	294/294 new modules, then the same isolated replay; 792 historical dependencies used from cache.	2402/2402 manifest entries.
Q394	860/860 new modules and isolated replay; three new endpoints. Historical dependencies mainly imported.	10157/10157 manifest entries.
Q395	1654/1654 new modules and isolated replay; 3228 historical dependencies used from cache.	24532/24532 entries; ZIP verified.

These figures measure different scopes: a Lake job, a direct source, an audited declaration, and a manifest entry are not interchangeable. Successful ZIP extraction is not compilation, and importing a cache is not a global rebuild.

Unfinished experiments remain separate from production. Q395 does not adopt its experimental shared audit; a partial performance measurement justifies neither an announced reduction in total cost nor removing a required check.

HEAD remains reported at 433e29e; local modifications and pre-existing lakefile additions are preserved. “No permanent Git operation” does not mean “pristine directory.” OVERRUN timeboxes and authorized continuations remain distinct from closeouts within the window.

Explicitly separated historical exception: the structural Q343 replay uses `Lean.ofReduceBool`; it does not supply the semantic proofs of the terminal campaigns. Empty scans are reported for their source scope, not extended retroactively to every experimental file.

DETAILED CHRONOLOGY

Milestone record 341-350

The “scope” column preserves each milestone's limits at its checkpoint, noting continuations that later removed them.

Milestone / theme	Result or role	Scope / status
Q341 Zero ordinate	TS341 closes NoZeroOrdinateInTruncation through the real eta route. [S1]	Proved; permanent result carried over from Volume I.
Q342 Interface audit	Sign → zero, disjointness → lower bound, upper bound + saturation → count. [S2]	Indirect traceability: connections reconstructed in Q343.
Q343 H = 1000 pilot	649 candidate brackets; Hardy/xi; count assembly under a semantic family and two Turing bounds. [S2]	Transport theorems proved; external FLINT computations, signs open.
Q344 First exact bracket	Rationals, serialized boxes, membership obligations, and sign transport. [S2]	Historical sources present; sign leaves still conditional at this stage.
Q345 FLINT conventions	Earlier convention analysis, mentioned in Q346. [S2]	Announced package and manifest absent according to Q346; no verdict reconstructed.
Q346 Hardy phase	Exponentiated phase independent of the valid logarithm; documented formula equals the Hardy carrier. [S2]	Bridge B closed; execution semantics and Bridge A open.
Q347 Rational checker	Soundness of atom/rational/neg/add/sub/mul and the AnalyticLeaf boundary. [S3]	Retry: kernel compiled outside the timebox; analytic atoms missing.
Q348 Theta-Mellin reduction	Exact identity, polar correction, theta tail, quadrature targets; repaired Q347 binding. [S3]	Analysis reduced; signs at 14 and 15 still conditional.
Q349 Mellin folding	Exponential substitution, integrability, and reduction to [0,2] plus a tail. [S4]	Semantic sources present and reused later.
Q350 Explicit tail	The true tail is bounded by 64/70336337001, hence by 10^{-9} . [S4]	Analytic bound available; compact sum not evaluated.

References and materials: page 28.

DETAILED CHRONOLOGY

Milestone record 351-360

The “scope” column preserves each milestone's limits at its checkpoint, noting continuations that later removed them.

Milestone / theme	Result or role	Scope / status
Q351 Trapezoidal rule	Error $\leq b-a ^3 M/(12N^2)$ under C^2 and a second-derivative bound. [S4]	Generic theorem; hypotheses still needed for theta.
Q352 Second derivative	Folded integrand C^2 ; second derivative ≤ 25 for $ t \leq 15$; specialization $N = 8192$. [S4]	Error certified; sum computation still open.
Q353 Transcendental traces	Register invariant and interpreter soundness; initial Taylor tools. [S4]	Kernel compiled; monolithic concrete replay not obtained.
Q354 Flattened trace	Checked dynamic registers, short blocks, Taylor cosine, and a real node at $u = 1$. [S4]	One node certified for 14 and 15; not the entire integral.
Q355 Interval kernels	References to flat kernels in the Q356 and Q361 dependencies. [S4, S5]	Indirect documentation; standalone report not found.
Q356 16-panel pilot	17 nodes, signed dyadic arithmetic, and a uniform theta majorant. [S4]	Standalone pilot closed; nominal Q349 binding not replayed at this checkpoint.
Q357 128-panel block	129 nodes bound to the Q349 integrand; balanced aggregation with exact endpoint weights. [S4]	First canonical block closed.
Q358 Sum and robust signs	8192 panels; theta-Mellin leaves; Hardy $Z(14) < 0 < \text{Hardy } Z(15)$. [S4, S12]	Genuine zero in (14,15); not the first serialized bracket.
Q359 Riemann-Siegel diagnostic	Derivative-capacity audit: only indices 0-9 in the initial experiment. [S5]	According to Q361: $K \leq 3$ usable; $K = 4-30$ incomplete.
Q360 Endpoint routing	Numerical census and reference envelope for order and endpoint selection. [S5, S10]	External diagnostic cited by later reports; not a sign proof.

References and materials: page 28.

DETAILED CHRONOLOGY

Milestone record 361-370

The “scope” column preserves each milestone's limits at its checkpoint, noting continuations that later removed them.

Milestone / theme	Result or role	Scope / status
Q361 Parameters and analytic gap	a, N, p, and positivity of envelopes; dependencies of the exact formula identified. [S5]	Parameters proved; decomposition and analytic remainder absent.
Q362 Primary definition of F	Corrected source quotient; half-integers as simple denominator zeros; first common zero. [S5]	Partial progress; entire extension not constructed.
Q363 Phases at half-integers	Reduction modulo 4 and numerator vanishing at every half-integer. [S5]	Generic common zero proved.
Q364 Entire extension	Local dslope model; global extension; agreement with the quotient and justified derivatives. [S5]	Entire function F and derivative interface closed.
Q365 Exact coefficients	Regular and top recurrences, support, Ck, finite sum, and indices $\leq 3K$. [S5]	Coefficient layer closed; no zeta transport at this stage.
Q366 Independent remainder	Initial provenance stop; Q366-R continuation: sourceRgK and sourceRSRemainder integrals. [S5, S6]	Definitions subsequently available; decomposition still conditional.
Q367 Majorant transport	Algebraic passage from a raw bound to the Hardy normalization factor. [S6]	Conditional on a genuine raw bound.
Q368 Riemann-Siegel certificate	Rational enlargement of a finite box by a radius; checker soundness. [S6]	Generic soundness; no concrete target obtained this way.
Q369 Admissible contours	Geometry, poles, branch, inner integrability, and Cauchy rectangles. [S6]	Fubini and infinite-shift gates still explicit.
Q370 Separable majorants	Gaussian exponent, factorizations, majorant integrability, and cosine coercivity. [S6]	Harmonic gap still open then; cosine completed in continuation.

References and materials: page 28.

DETAILED CHRONOLOGY

Milestone record 371-380

The “scope” column preserves each milestone's limits at its checkpoint, noting continuations that later removed them.

Milestone / theme	Result or role	Scope / status
Q371 Planned production endpoint	Canonical target placed after the Q369-Q370 analytic gates. [S6]	Not attempted in the RS3H-II report.
Q372 Planned first RS bracket	Target pair of signs via the new Riemann-Siegel route. [S6]	Not attempted in the RS3H-II report.
Q373 Harmonic gap	Boundaries, positivity, compactness, and tail; SourceFRealGap witness constructed. [S6]	Product domination and Fubini without a residual gap premise.
Q374 Source assembly	Semantic Fubini and audit of 2011/2024 conventions; assembly steps. [S6, S7]	Source formula open in the first report; closed through Q379-Q380.
Q375 Remainder bound	Remainder norm dominated by majorant integrals, then closed-form variants. [S6, S7]	Not the generic Arias theorem; sharp quantification kept separate.
Q376 Sides and contour shift	Transverse decay, truncated Cauchy, and deformation of the primary Taylor difference. [S7]	Binding to sourceRgK on the final line still missing.
Q377 Parameter analyticity	Transport domain and continuation; equality interface on the final line. [S7]	Independent local identity still required.
Q378 Circle to line	Cauchy on an annulus, outer contour, and inner Blaschke parametrization. [S7]	Global inner angular lift remains to be completed.
Q379 Complete inner transport	Degree 1, increment 2π , oriented substitution, germ, then continuation. [S7]	Remainder and primary correction connected; independent representation of R still awaited.
Q380 Representation of R	Primary contour, one-pole jump, telescoping, substitution, and source Riemann-Siegel formula. [S7]	Source formula closed; auxiliary-zeta relation still open.

References and materials: page 28.

DETAILED CHRONOLOGY

Milestone record 381-390

The “scope” column preserves each milestone's limits at its checkpoint, noting continuations that later removed them.

Milestone / theme	Result or role	Scope / status
Q381 Reflection and entirety	ΓR , reflected contour, branch, orientation, entire R , and real critical pair. [S8]	Hardy connector conditional on the auxiliary-zeta identity.
Q382 Primary Mordell formula	Closed form, deformations, decay, and exact oblique Gaussian. [S8]	Mordell closed; Mellin transport to the auxiliary pair absent.
Q383 Theta and Mellin	Regularization at zero, integrability, sum-integral interchange, and Gaussian Mellin = completed zeta. [S8]	Primary transport to theta still open.
Q384 Regular transport	Canonical geometry; identity on a regular strip; positive odd Gamma exceptions. [S8, S9]	Old global seed retains an exceptional-point premise.
Q385 Distinguishing two functions	Zeros of totalized Gamma and of the entire auxiliary at negative even points. [S9]	Does not prove zeros of the completed theta expression.
Q386 Regular values	Regularity and exact representation of the completed value by a half-integral. [S9]	No vanishing, nonvanishing, or closed form obtained.
Q387 Conditional obstruction	Reduction at positive odd points; reindexing; nonzero value $\Rightarrow$ old seed fails. [S9]	The nonvanishing hypothesis is not constructed.
Q388 Constant mode separated	Integrability, line decomposition, and exact residual-cancellation criterion. [S9]	Residual integral not evaluated.
Q389 The entire critical line	Upper/lower regular continuation, limit at $t = 0$, Hardy transport, and exact decomposition. [S9, S10]	Analytic chain closed for this target; concrete endpoint still open.
Q390 Canonical point and envelope	t_0 , $K = 63$, $\text{ell} = 10$, central regime, cosine $M = 1$, $\text{Arias} \leq 10^{-14}$. [S10]	True remainder transport and atoms still open.

References and materials: page 28.

DETAILED CHRONOLOGY

Milestone record 391-396

The “scope” column preserves each milestone's limits at its checkpoint, noting continuations that later removed them.

Milestone / theme	Result or role	Scope / status
Q391 Effective pointwise remainder	Independent bound of 2.5×10^{-30} ; RequestedAriasRemainderTransport inhabited without premises. [S10]	Requested remainder closed; finite atoms next.
Q392 Six atoms out of eight	Gamma/Dirichlet, then prefactor; certified boxes and replay with remaining atoms explicit. [S11]	Two finite-correction components still open.
Q393 Eight atoms and an endpoint	190 jets, 64 rows, and 3072 contributions; memberships, positivity, and a concrete AnalyticLeaf. [S11]	Right endpoint 415 closed; pair and family still to complete.
Q394 Two historical brackets	Three new points; rows 414-415; distinct zeros; uniform radius extended to eleven spans. [S12]	Subfamily of 2; lower bound 3 with (14,15), separately.
Q395 Six historical brackets	Eight new endpoints; rows 410-413 added; exact cardinality 6 and multiplicity lower bound ≥ 6 . [S12]	405-409 open; lower bound ≥ 7 with (14,15); Bridge A open.
Q396 Continuation mandate	Reused certified inverse, a complete experiment, then five new pairs targeted in the covered range. [S13]	Results not supplied. No Q396 pair is added to this assessment.

Handoff to a new conversation: start from the final Q395 result and the exact subfamily {409,410,411,412,413,414}. Retain Q391, the eight Q393 atoms, and the uniform Q394 bound as established within their scopes. Request the Q396 report before adding any of its results to this chronology.

OPEN statuses in older documents express the frontier at their date. They must neither erase a later proof nor be replaced by a retroactive closeout of the entire sprint.

CONTINUATION GLOSSARY

Notation and objects to keep distinct

Object	Meaning in the chain
$s = 1/2 + it$	criticalPoint t, a point on the critical line of the zeta function.
normalizedHardyZ	The project's canonical real function, connected to the Gamma phase; the target of sign proofs.
$\Gamma(s)$	Mathlib's complex real-Gamma factor, corresponding to $\pi^{(-s/2)} \Gamma(s/2)$ at regular points. Its totalization at poles requires distinguishing its Lean value from analytic continuation.
sourceAuxiliaryR	R(s), independently defined by its primary contour integral.
sourceAuxiliaryPair	Sum of the completed auxiliary and its conjugated reflected term. Q389 connects it to zeta on the critical line.
riemannSiegelF	Q364's entire function obtained from the regularized quotient N/D. Its derivatives enter the Q365 coefficients.
sourceF	Logarithmic function from the 2024 source, used in the remainder. It is not riemannSiegelF.
sourceRSRemainder	Independent integral remainder; its representation and bounds are proved before the sign certificate.
Normalized jet of order n	Taylor coefficient, equal to the nth derivative divided by n!. The canonical assembly restores the required n! factors.
Atom / membership	Elementary real value of the finite expression / proof that this true value belongs to the proposed interval.
AnalyticLeaf	Object pairing a rational output with a membership proof for the true function at the true endpoint.
Bracket / ledger	Rational interval carrying a sign change / historical table of rationals. Row j corresponds to Lean index j−1.
Payload / saturation	Box and cap data / proved connection between local lower bounds, the global upper bound, and the exact sum.

Here, “fail-closed” means that missing results remain explicit obligations: no external datum, default value, or unconstructed hypothesis is presented as a terminal proof.

DOCUMENTARY EVIDENCE

Sources and reproduction landmarks

Internal references to the supplied materials. Lean sources specify types and dependencies; logs and attestations describe the checks performed during each campaign.

Ref.	Main materials consulted
S1	TS1-340-Horizon-Goldbach-Program-Synthesis(1).pdf, especially pages 18-22 and the reproducibility appendix.
S2	Q343_REPORT.md; Q346_REPORT.md; Q344 sources and Q343 bindings archived in recent packages.
S3	Q347_REPORT.md, initial version and Retry archive; Q348_REPORT.md and certificate-kernel sources.
S4	Q349-Q351 sources; Q352-Q354 and Q356-Q357 reports; Q358Probe/TerminalSigns.lean and its consumers.
S5	Q361-Q366 reports and specifications; Q355/Q359/Q360 information retained in Q361.
S6	RS3H2/RS3H3_REPORT.md; Q366-R, Q367-Q368; Q369, Q370, Q373, Q374, and Q375 reports.
S7	Q376-Q380 reports and RS4H-RS8H archives; continuations of the Q374 source formula.
S8	RS9H-RS11H reports, Q381 specification, and Q384 transport; Mordell and theta-Mellin theorems.
S9	Q384-Q389 reports and sources; Q385-Q388 exceptional values; Q389 continuation specification.
S10	Q390_REPORT.md; Q391_CLOSURE_REPORT.md; RS19H CLOSED archive and closeout attestation.
S11	Q392 checkpoints and prefactor correction; Q393_REPORT.md, COEFFICIENT_SPEC, and BRIDGE_A_CONSUMERS.
S12	Q394-Q395 reports, statuses, and contract maps; final HistoricalClosedFamily source; Q395 documentary correction described in the closeout.
S13	Q396_RS24H_MANDAT_EXCLUSIF.md: authorization and work plan, without an attached results report.

Pinned environment. Lean 4.15.0; Mathlib 9837ca9d65d9de6fad1ef4381750ca688774e608.

Reference HEAD: 433e29eff1d9da2bc5937bb82bf5df3a2d59c08f.

Archives contain sources and evidence according to their inventories; required external caches remain specified in the replay procedures.

Latest closeout package: RS23H / Q395.

Archive SHA-256:

181f59421bd1e4867a83527aae4f8e7292b78ee52c2077ae8bf37ec82826531d

Manifest SHA-256:

ddf656fc8d30502739cdfcf89e34e3ec5fee97f3c45426383cd2b7f8b33f2f89

Documentary synthesis completed September 9, 2026. Q396: results not supplied.